Human Evaluation Study

Context-Aware Empathetic Text Generation for Mental Health Support

Build instructions, complete item set, and analysis plan.

1. What this study is for

The study has two goals, and the second matters more than the first.

- Goal 1 - Independent evidence for the main claim: do real people prefer our system's replies over the baselines?
- Goal 2 (the important one) - Validate the AI judges. Every judge-based result in the results chapter currently rests on three AI models. If human raters agree with those judges, that entire evidence base becomes defensible. This is why every conversation below is drawn from the 64 conversations that all three AI judges scored.

2. Design decisions and the evidence behind them

Decision	What we do	Why
Comparison format	Forced choice between replies shown side by side	Both human-evaluation papers in our reading list (COLING 2025 main.355 and main.598) use A/B preference rather than solo rating. main.598 gives the reason: pairwise choice 'accounts for individual variations among annotators'.
Replies per item	3	A deviation from the papers' 2, forced by rater budget. With ~12 items a pairwise design would test only one comparison. Three-at-once tests every comparison on every item.
Dimensions	Empathy only	Fatigue control. main.355 used 4 aspects and main.598 used 3, but with only 3 and 7 expert annotators. We trade aspect breadth for rater count.
Item count	13 (11 scored + 2 attention checks)	Keeps completion near 13-15 minutes. Past roughly 15 minutes volunteers begin answering carelessly.
Item selection	Random, fixed seed, one conversation per emotion	Removes any possibility of cherry-picking favourable examples - the easiest attack on a human study.
Student situations	3 items fixed to student scenarios	Most raters are students. A study whose situations are all about spouses, children and jobs asks them to judge empathy for lives they have not lived. Only 4 of the 64 judged conversations are

		student-related and none from a student's own point of view, so these three come from the wider 300-conversation test set and are not AI-judge-scored - see the limitation on the last page.
Reply ordering	Randomised independently for every item	Controls position bias. If our system always appeared first the result would be worthless.
Attention checks	2 items contain an obviously poor reply	Directly addresses the risk that bored raters click randomly. Anyone selecting the decoy is excluded from analysis.
Which items become checks	The conversations where the AI judges scored the decoy lowest	A check only works if the decoy is unmistakably poor. If it is merely mediocre, a careful rater may genuinely prefer it and then be wrongly excluded - worse than having no check. The two chosen here scored 2.0 and 2.7 out of 10 from the judge panel, against a decoy average of about 3.5.

Systems being compared:

Rater sees	Actually is	Why it is in the study
A, B or C	Our system (two-stage, Phi-4-mini)	The system the thesis claims.
A, B or C	Single model, same size, no detector	The core claim: does the decoupled design beat a monolithic model of equal size?
A, B or C	Single bigger model, no detector	The honest hard test. The AI judges say we lose this one; including it anyway is what makes the study credible.
A, B or C (2 items only)	Fine-tuned monolithic model	Attention-check decoy. Produces ~200-word replies full of questions and jokes.

3. How to build the Google Form

Settings to fix before adding any questions:

- Do NOT collect email addresses - the study is anonymous.
- Turn OFF 'Shuffle question order'. Reply order is already randomised inside each item; shuffling items as well makes the answer key harder to track.
- Mark every item Required, so there are no partial responses to clean up.

Never write a model's name anywhere in the form. Raters see only Response A, Response B and Response C.

Section 1 - Introduction and consent (paste this text):

Thank you for helping with this study.

You will read 13 short messages from people describing something they are going through. For each one, you will see three possible replies. Your task is simply to choose which reply you feel responds best to that person's feelings.

There are no right or wrong answers - we want your honest impression.

This takes about 12-15 minutes. Your responses are completely anonymous; we do not collect your name or email. Some messages describe sad or stressful situations. You may stop at any time.

By continuing, you agree to take part.

Section 2 - Two background questions (optional but useful):

- Are you a: Student / Faculty member / Other
- Have you studied psychology or counselling? Yes / No

These let you report who your raters were, and check whether faculty and students answered differently.

Section 3 - The 13 items. Each is laid out exactly like this:

[Title] Situation 1 of 13
[Description] Someone wrote: "...the person's message..."

Response A: ...
Response B: ...
Response C: ...

[Question - multiple choice - required]
Which reply best understands and responds to how this person is feeling?
Response A / Response B / Response C

Section 4 - Closing (optional free text):

"If anything stood out to you about the replies you read, feel free to tell us here." (Paragraph answer, not required.)

4. The question wording

Use exactly this wording on every item, with no variation:

Which reply best understands and responds to how this person is feeling?

Options: Response A / Response B / Response C. Required.

There is deliberately no 'they are the same' option. With three replies on screen a tie option absorbs a large share of answers and weakens the result; the papers used Tie only in two-way comparisons.

5. The 13 items - copy these into the form

Reply order below is already randomised. Paste exactly as shown: do not reorder, do not relabel, and do not include the word counts.

Situation 1 of 13

Someone wrote: *"when i was younger I was really getting into graphic design and really wanted a new Apple computer. I knew my parents couldnt afford it, but they somehow made it happen and surprised me on Christmas.."*

Response A: Oh, that's such an exciting memory! It must have been magical for you to get something so special. That kind of support from family is truly meaningful. Would you like to talk more about those early days of graphic design or maybe share anything else about that time?

Response B: I can only imagine how overwhelming and amazing that must have felt, knowing your parents went above and beyond to surprise you with something you truly loved. It's such a special memory of love and support from your family. Knowing they prioritized making you happy like that is incredibly heartwarming. It sounds like those moments of surprise and support shaped your journey in graphic design. How do you feel looking back on that time now?

Response C: Wow, that's incredibly surprising and heartwarming! It's amazing how unexpected gifts can bring such joy. It's comforting to know that loved ones can make such wonderful sacrifices for us. It must have been an incredibly special moment for you. How did you feel when you saw the gift?

Q: Which reply best understands and responds to how this person is feeling?

Situation 2 of 13

Someone wrote: *"Have you ever played and old video game called Everquest?"*

Response A: I can see how Everquest brings back fond memories for you. It's amazing how old games can feel so comforting, isn't it? Remembering those moments with friends must be so nostalgic and warm. Have you thought about diving back into Everquest with a friend, perhaps rekindling some old memories?

Response B: Absolutely, I've heard of that! It sounds like an intriguing adventure. What do you enjoy most about it?

Response C: Yes! I have never really done much with it but I remember it being cool when I was younger. Have you? What did you think of it? Was it fun? Or boring? I am curious. I know people who still play it today! It sounds like something they might be able to enjoy even if they haven't in years! Maybe we can talk about it sometime soon! lol. Do you play any games now or just nostalgia? Just wondering. Is there anything else you want to ask me about too? :) Good luck on your exam tomorrow! You've got this! :D. And I hope all goes well

for you! Thank you so much for talking with me tonight! I appreciate it! I'm glad I could help at least a little bit! :) If you need anything else please let me know! I'll try my best to help! :) Have an awesome day! Take care! :) Peace out! :) I love you guys! :) Bye-bye! :) Have a good one! ;] I miss you all! :) See you later! ;] I'll see you again soon! :) I promise! I don't forget you! :) Talk to you soon!

Q: Which reply best understands and responds to how this person is feeling?

Situation 3 of 13

Someone wrote: *"I am visiting my friend this week."*

Response A: It sounds like you're looking forward to spending time with your friend and catching up. It's wonderful to have those moments of connection and support. Knowing you'll have that time together can be really comforting. How are you planning to make the most of your visit?

Response B: I understand that you're excited about visiting your friend. It's wonderful to have the opportunity to catch up and create new memories together. Your visit sounds like a perfect chance to relax and enjoy each other's company. How are you feeling right now?

Response C: That sounds like an exciting time for both of you! I'm sure it's going to be great. Are there any specific things that make you nervous about visiting? Does anything excite you most about spending time with your friend?

Q: Which reply best understands and responds to how this person is feeling?

Situation 4 of 13

Someone wrote: *"If my upstairs neighbor wakes me at 3am again I think I'll explode! I get that she has to go to work, but why not get organized the night before for goodness sake?!"*

Response A: I understand how frustrating it must be for you when your upstairs neighbor disturbs your sleep. It's understandable that this could be stressful. Remember, waking up early can be challenging even under normal circumstances. What specifically do you need from her so that you can manage this better?

Response B: I can really understand how frustrating and overwhelming it must be to have your sleep disrupted like that. It sounds like you're dealing with a lot of stress and maybe even some anger about the situation. Knowing that she has responsibilities is important, but it's understandable to feel like things could be handled better. Does talking to her about it seem like a good idea to you?

Response C: I hear how frustrating it is to be woken up by your upstairs neighbor at 3am. It's perfectly understandable to feel annoyed when something that should be predictable

disrupts your sleep. Maybe try talking to her about setting up a more predictable routine? How do you feel about discussing this with her?

Q: Which reply best understands and responds to how this person is feeling?

Situation 5 of 13

Someone wrote: *"My Calculus final was no joke! I started studying hard two weeks before the test."*

Response A: Your dedication to studying Calculus really paid off, and it's fantastic to see how prepared you felt for your final. Your hard work and commitment truly deserve recognition. Remember, your knowledge and understanding of the subject matter are powerful tools. Is there anything else on your mind after this accomplishment?

Response B: I'm so glad to hear that you've been putting in the effort! That's truly admirable. You're doing great by taking control of your studies. What next steps do you plan to take after this exam?

Response C: I can only imagine how stressful those last few days must have been for you. It sounds like you put in a lot of effort to prepare, but it can still be overwhelming. You did everything you could to be ready. How do you think you did on the test?

Q: Which reply best understands and responds to how this person is feeling?

Situation 6 of 13

Someone wrote: *"My children recently started school again and it made me feel a bit apprehensive. They are growing up so quick, and I know it won't be long before they sadly move out of the house :("*

Response A: I understand how starting school again can bring a mix of excitement and apprehension, especially when thinking about your children growing up so fast and the bittersweet moments of them moving out. It's perfectly normal to feel a bit anxious about these changes. Remember, these are just stepping stones in their journey, and you are a constant source of love and support for them. How are you feeling about their upcoming school event?

Response B: I understand that starting back at school can make parents like us feel anxious too. It's natural for us to miss our children when we don't see them regularly. But remember, they're growing up fast and moving on from home is part of becoming independent adults. Everything happens for a reason, even if we can’t always see why right now. How do you think about this situation?

Response C: I can really understand how bittersweet and a bit scary that can be. It's such a big transition for both you and the kids. Knowing they're growing and learning is amazing,

but it does bring a lot of emotions. How do you think they are adjusting to being back in school?

Q: Which reply best understands and responds to how this person is feeling?

Situation 7 of 13

Someone wrote: *"In college I was in a class with an older woman who stole my ideas for a paper."*

Response A: I'm so sorry that happened to you. It's hard when someone takes credit for work you put in. Just remember, you can always reach out for support from friends or professionals like counselors or mentors. The most important thing is that you stand strong and don't let this setback define you. What do you think about sharing more details about how you're handling it right now?

Response B: I can only imagine how hurt and frustrated you must feel. It's so unfair when others take credit for our hard work. You deserve recognition for your efforts. Did this experience change how you approach sharing your ideas in the future?

Response C: I hear you, and it's completely valid to feel annoyed in this situation. It's tough when someone in a position of authority takes credit for your work. Remember, your ideas and contributions are valuable and deserve recognition. Just know that this doesn't define you or your abilities. Keep focusing on your passion and dedication. Would you like to talk more about this?

Q: Which reply best understands and responds to how this person is feeling?

Situation 8 of 13

Someone wrote: *"I took a test for college the other day, it was basically our midterm."*

Response A: I can really understand how nerve-wracking that must have been, especially since it feels like so much is riding on it. It's great that you're taking steps to prepare. How do you think it went?

Response B: I can sense your excitement and anticipation from taking your college test. It's a significant milestone, and your hard work is paying off. Remember, each step forward brings you closer to your dream college. How are you feeling right now?

Response C: Your test must have felt pretty challenging, huh? I'm here to listen if you want to talk about it. Is there anything specific that's on your mind right now?

Q: Which reply best understands and responds to how this person is feeling?

Situation 9 of 13

Someone wrote: *"So I moved away from my parents not too long ago. It's been a big change."*

Response A: I can only imagine how overwhelming and bittersweet that must be for you. Leaving home can be so hard, but it's also a brave step towards independence. You're doing great by taking this step. How are you finding life on your own so far?

Response B: I hear how moving away from your parents must feel isolating and challenging. It's natural to miss their presence and feel the weight of this new distance. Remember, even though you're physically apart, the love and connection you share can transcend distance. How have you been feeling emotionally since the move?

Response C: I'm really sorry to hear about that move. Moving away can be tough, but it's also an exciting step into new opportunities. How do you find yourself adjusting to this change?

Q: Which reply best understands and responds to how this person is feeling?

Situation 10 of 13

Someone wrote: *"I made a bowl of cereal for breakfast and when i started eating i realized milk was spoiled!! I started gagging. it was so gross!"*

Response A: I understand that discovering spoiled milk in your breakfast cereal would be deeply upsetting and disgusting. It's completely normal to feel nauseated and revolted in this situation. Remember, it's okay to feel this way and to feel disgusted. The fresh morning air outside might help cleanse your senses and restore a bit of comfort. Would you like to talk about how you're feeling right now?

Response B: Oh my goodness, that must have been incredibly upsetting for you. It's understandable how you would be feeling upset about making such an error. What a tough situation. I'm here to support you through this. Is there anything specific you'd like to talk about or need guidance on?

Response C: I can really understand how upsetting that must be; seeing spoiled milk in your cereal can be pretty scary and gross. It's completely normal to feel nauseous and grossed out by that experience. You're safe though, and hopefully, you'll be able to move on from this. Did the rest of your breakfast go okay after that?

Q: Which reply best understands and responds to how this person is feeling?

Situation 11 of 13

Someone wrote: *"My dog disappeared while I was away at college."*

Response A: I'm so sorry about that. Losing a pet can be incredibly heartbreaking. Please know that there is no time limit on grief; it's okay to grieve for days or even months after losing someone. How are you finding comfort during this difficult time?

Response B: I'm so sorry to hear that your dog went missing; it must be incredibly worrying and heartbreakingly sad for you. Losing a pet can feel like a huge loss, but know that you're not alone in this pain. Have you had any updates since you found out?

Response C: I hear how devastating this must be for you. It's completely understandable to feel overwhelmed and heartbroken when someone so close to you vanishes. It's okay to grieve the absence of your beloved pet. Remember, their spirit and memories will always bring comfort and love to your heart. How are you holding up?

Q: Which reply best understands and responds to how this person is feeling?

Situation 12 of 13

Someone wrote: *"i am impressed that video games grow so much every year"*

Response A: That's truly impressive! It shows the incredible creativity and innovation that game developers bring to our screens each year. And for many people, gaming can be a fantastic escape into worlds of imagination and adventure. Is there anything specific about gaming or its evolution that interests you?

Response B: I agree. I have seen some amazing games recently! What do you like playing? I can't wait for this one new game coming out though... it looks awesome! I'll have to get my hands on it as soon as possible! I hope you enjoy it! Do you play games often or only when you're bored? I know people who will never play any games ever unless they feel compelled to. How about you? I'm not sure how many games I play in a month but I think most of them are from time to time rather than constantly. I don't really like spending all day gaming. I prefer being outside and enjoying nature. I think that was what brought me back into video games again after having been away for quite a while. It made me realize I was missing out by not going out more. I also realized that I had a lot of free time to spend on things other than just sitting around doing nothing. So now I'm trying to make use of it and go out more. I think it has helped me be happier with life overall. Video games aren't my favorite thing anymore because I've found a whole bunch of other things to enjoy instead. But if I want to watch TV or something else I'd

Response C: Your admiration for the growth of video games is truly inspiring. It's fascinating how these games continue to innovate and captivate players worldwide. It's comforting to know that creators are pushing boundaries, leading to such impressive advancements. Your excitement about this evolution must be quite rewarding. What specific game or development recently caught your attention?

Q: Which reply best understands and responds to how this person is feeling?

Situation 13 of 13

Someone wrote: *"Been having a good day, relaxing on my couch watching Netflix. I got a nice cup of hot chocolate and my cat curled up in my lap."*

Response A: It sounds like you're having a wonderful, relaxing day. The comfort of your hot chocolate, Netflix, and your cat's warmth in your lap really seems to be putting you at ease. Remember, moments like these are perfect for simply enjoying the peace around you. How are you feeling right now?

Response B: That sounds incredibly peaceful! It's wonderful that you're finding moments like this to relax and enjoy yourself. Your well-deserved break feels so much better than anything else.

Response C: It sounds like you're really enjoying some peaceful moments to unwind. It's wonderful to have those small, cozy pleasures that can make a big difference. You deserve this downtime to recharge. How long do you usually spend watching Netflix before you decide to do something else?

Q: Which reply best understands and responds to how this person is feeling?

6. How to analyse the responses

Run these in order. Step 1 comes first, because it decides whose data counts.

- Step 1 - Exclusion. Drop any rater who selected the decoy reply on either attention-check item. Report how many were dropped; this is evidence of quality control, not something to hide.
- Step 2 - Selection rate per system. What share of all choices went to each system? Chance is 33.3%. Test our system against chance with a binomial test.
- Step 3 - Head to head. For each rater, count how often they chose our system versus each baseline, then compare those counts across raters with a Wilcoxon signed-rank test. This is the number that answers the core claim.
- Step 4 - Rater agreement. Compute Fleiss' kappa across raters, to show people were not guessing. Report it even if it is low - a low kappa on a subjective task is a normal and publishable finding.
- Step 5 - The important one. For each item compare the humans' majority choice against whichever system the AI judges scored highest, and report the agreement rate. This is what justifies every AI-judge result in the thesis.

Step 5 needs the AI-judge scores for the two-stage system, which are still being collected. Everything else can run the moment responses arrive.

7. Limitations to report honestly

- Raters are classmates, friends and faculty - not clinicians, and not people actually seeking mental-health support.
- 13 conversations is a small item set. The comparison papers used 100 and 200 conversations, though with far fewer raters (3 and 7) than we expect to recruit.
- Raters may know the authors, which can bias them toward a favourable answer. Blinding the system names is the protection against this, which is why no model name may appear anywhere in the form.
- Only empathy was measured. Fluency, coherence and informativity were not.
- Situations 5, 7, 8 are student scenarios taken from the wider test set, and were not scored by the AI judges. They therefore count towards the preference result but not towards the human-vs-AI-judge agreement analysis, which runs on the 8 scored items that were judged. That is enough for the agreement figure, but it should be stated rather than left for a reader to discover.

Emotions covered by the item set:

angry, anticipating, anxious, content, disgusted, faithful, furious, impressed, joyful, lonely, nostalgic, prepared, sad.

13 of the 32 EmpatheticDialogues emotions, one conversation each. Four families central to this application (sad, lonely, angry, anxious) were guaranteed a place; the remaining items were drawn at random. Which conversation represents each emotion was random in every case, forced or not, so the set covers what a reader would expect without any scope to hand-pick favourable examples.

8. Answer key - which reply is ours

DO NOT show this page to raters, and do not paste it into the form. It exists so you can score the responses after collection.

Cross-check against ANSWER_KEY_do_not_share_with_raters.csv, which additionally records what the other two responses were on every item.

Situation 1: Our model reply: Response C

Situation 2: Our model reply: Response A [attention check - Response C is the decoy; exclude any rater who chose it]

Situation 3: Our model reply: Response B

Situation 4: Our model reply: Response C

Situation 5: Our model reply: Response A

Situation 6: Our model reply: Response A

Situation 7: Our model reply: Response C

Situation 8: Our model reply: Response B

Situation 9: Our model reply: Response B

Situation 10: Our model reply: Response A

Situation 11: Our model reply: Response C

Situation 12: Our model reply: Response C [attention check - Response B is the decoy; exclude any rater who chose it]

Situation 13: Our model reply: Response A

Full breakdown of every position:

Situation	Emotion	Response A	Response B	Response C
1	joyful	Single model, same size (no detector)	Single bigger model (no detector)	Our system (two-stage, Phi-4-mini)
2	nostalgic	Our system (two-stage, Phi-4-mini)	Single model, same size (no detector)	Fine-tuned monolithic model [DECOY]
3	faithful	Single bigger model (no detector)	Our system (two-stage, Phi-4-mini)	Single model, same size (no detector)
4	angry	Single model, same size (no detector)	Single bigger model (no detector)	Our system (two-stage, Phi-4-mini)
5	prepared	Our system (two-stage, Phi-4-mini)	Single model, same size (no detector)	Single bigger model (no detector)
6	anxious	Our system (two-stage, Phi-4-mini)	Single model, same size (no detector)	Single bigger model (no detector)
7	furious	Single model, same size (no detector)	Single bigger model (no detector)	Our system (two-stage, Phi-4-mini)

8	anticipating	Single bigger model (no detector)	Our system (two-stage, Phi-4-mini)	Single model, same size (no detector)
9	lonely	Single bigger model (no detector)	Our system (two-stage, Phi-4-mini)	Single model, same size (no detector)
10	disgusted	Our system (two-stage, Phi-4-mini)	Single model, same size (no detector)	Single bigger model (no detector)
11	sad	Single model, same size (no detector)	Single bigger model (no detector)	Our system (two-stage, Phi-4-mini)
12	impressed	Single model, same size (no detector)	Fine-tuned monolithic model [DECOY]	Our system (two-stage, Phi-4-mini)
13	content	Our system (two-stage, Phi-4-mini)	Single model, same size (no detector)	Single bigger model (no detector)